

QUESTIONS AND ANSWERS

Q Why is the sea salty?

A The sea's salinity comes from salt washed out of the Earth's rock, sand, and soils by rainwater then carried in streams and rivers to the sea. This has been happening for millions of years, building up the sea's concentration of salt. Sodium chloride, or common salt (the kind we put on our food), makes up around 85% of the minerals in the sea.



Waves forming out at sea

Q What causes waves?

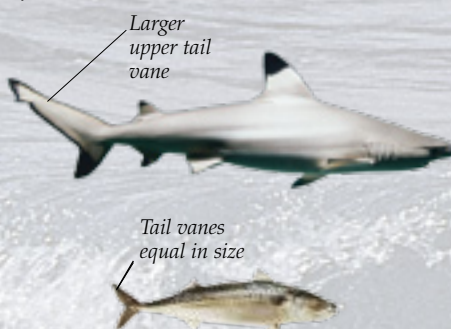
A Most waves are created by the wind blowing across the open ocean and causing the surface water to ripple. If the wind continues to blow, the ripples grow larger and turn into waves. The height of waves depends on the strength and duration of the wind causing them and how far they have been "pushed" across the ocean.

Q Why is the sea blue?

A Blue light from sunlight (which is made up of all the colors of the rainbow) is the least absorbed by clear seawater. When sunlight enters the water, the blue light is scattered and some is reflected back to the surface, making the sea look blue.

Q How many types of fish are there?

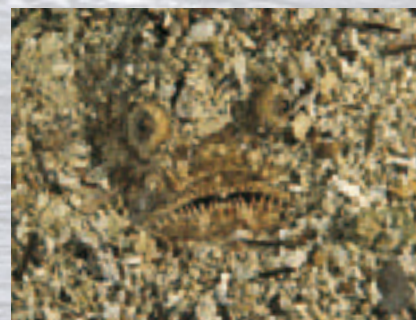
A There are around 25,000 different species of marine and freshwater fish. Of this number, around 24,000 are bony fish, 1,100 are cartilaginous or gristly fish (such as sharks and rays), and about 100 are hagfish and lampreys—the so-called jawless fishes.



Cartilaginous fish—blacktip reef shark (top) and bony fish—mackerel (above)

Q What is the difference between bony and cartilaginous fish?

A As their name suggests, bony fish have bony skeletons whereas sharks and rays have skeletons made of cartilage (gristle). In addition, a bony fish has a gas-filled swim bladder to control buoyancy. This can be adjusted to make the fish weightless in water so it can rest or remain motionless. Gristly fish, however, will visually sink if they don't keep moving. Other characteristic differences are that bony fish have tail vanes of equal size and protective gill flaps, whereas most sharks have larger upper tail vanes than lower ones, and gill slits without flaps.



Stargazer hiding in gravel on the seabed

Q How do fish hide from predators in the open sea?

A Many fish that live near the sea's surface have dark backs and paler underbellies. This countershading camouflages them from above and below. Fish that live on the seabed are often colored and patterned to blend in with their surroundings.

Record Breakers



LARGEST SEA CREATURE

The blue whale is the world's largest animal, at up to 100 ft (30 m) long and as much as 165 tons (150 metric tons) in weight.



BIGGEST FISH

The whale shark can grow up to 42 ft (12.5 m) long and weigh up to 22 tons (20 metric tons).



SMALLEST FISH

The adult Marshall Islands dwarf goby is just $\frac{1}{2}$ in (6 mm) from nose to tail.



HEAVIEST BONY FISH

The ocean sunfish, or *Mola mola*, can weigh up to 2 tons.



FASTEST FISH

The sailfish can reach speeds of up to 68 mph (109 km/h)—faster than a cheetah can run.



Whale shark